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PEOPLE
MARIE CURIE ACTIONS

Marie Curie CareerIntegration Grants (CIG)

Call: FP7-PEOPLE-2012-CIG

PART B

PROPOSAL

“CHEMiRNA”

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B1. SCIENTIFIC AND TECHNOLOGICAL QUALITY

Project Title: *CHEMiRNA: Chemical-based Platforms for micro-RNA Detection. Towards Novel OncomiR Assays*

B 1.1. Scientific and technological Quality, including any interdisciplinary and multidisciplinary aspects of the proposal

Project Aims: The aim of the proposed research is to design novel reagents and tools for the detection and quantification of microRNAs (miRNAs) with a particular focus on miRNA species involved in solid tumours and hematologic cancers. The proposed project will thus provide a new chemical toolbox which will allow the rapid detection of miRNAs involved in cancers. This proposal intends to take advantage of the novel dynamic chemistry approach to profile directly nucleic acids and develop based on it novel tools to provide simple, sensitive, rapid, multiplexed assays to profile miRNAs. The longer-term vision is that this dynamic chemistry approach will enable clinicians to be able to rapidly and accurately diagnose and prognosis cancer by combining the precision of the high affinity molecular probes and novel set of biomarkers based on miRNAs. Therefore, the proposed project is positioned at the interface of chemistry, biology and pre-clinical medicine with three main aims:

(i) *Design, synthesis and validation of chemical reagent sets for novel cost-effective miRNA profiling assays.*

(ii) *Development of microsphere miRNA-capture devices that will allow monitoring of miRNAs directly from body fluids.*

(iii) *Development of microarray-based detection method “miOncochip”, for oncomiRs detection.*

Background of the State of the Art of Research in the Field

During the past decade microRNAs (a novel species of non-coding RNAs with gene regulatory function) have been linked virtually to all known physiological and pathological processes, including cancer.¹⁻³ In the same way as certain key protein-coding genes, miRNAs can be deregulated in cancer, in which they can function as a group to mark differentiation states (cancer classifiers) or individually as real oncogenes or tumour suppressors (cancer genes).⁴ Understanding the role of miRNA in biological processes is thus key to investigate cancer phenotypes, diagnose and classified cancer types in a much stratified manner and also to develop new therapies which either target or mimics miRNAs. Components of the miRNA machinery and miRNAs themselves are involved in many cellular processes that are altered in cancer, such as differentiation, proliferation and apoptosis.⁵ *In vitro* studies have showed evidence implicating miRNAs such as miR-21 and miR-10 in cancer.⁶ For instance, in miRNA profiling experiments, miR-21 is found to be overexpressed in most of the tumour types examined so far, including neuroblastoma, glioblastoma, colorectal, lung, breast, pancreas, leukaemia and lymphoma.⁷⁻¹⁰ Recently, Medina *et al* have studied the profile of miRNA expression in 540 human samples for six tumour types, miR-21 was the only miRNA upregulated in all types.^{6, 11} To analyse miRNA scientist adapted tools which are use for the majority of DNA genotyping methods at present relying upon DNA polymerase chain reaction (PCR) for the exponential amplification of DNA target molecules. Therefore, most of the efforts to do miRNA analysis have been based on PCR technologies to create cDNA from miRNA. However this approach is particularly challenging to miRNA short nature (22-24 nucleotides). This means that standard protocols for miRNA analysis are long and still inefficient (to date a few reports comparing different platforms cannot find very good reproducibility between them), including a ligation step to elongate the short RNA sequences reverse transcription and PCR to provide a product, cDNA, that can be analysed either via qRT-PCR or arrays (surfaces or beads).¹²⁻¹³ Due to the striking role of miRNA in cancer - it is commonly accepted that the widespread alteration in the expression of miRNAs is an omnipresent feature of cancer-most of the biological assay providers have entered into this market. Amongst the one

offering solution approaches we can find Life Technologies with their TaqMan approach, classically used for genotyping, and Exiqon with their LNA technology to create miRNA specific primers. These technologies cannot be run in a multiplexed manner and their robustness in terms of quantification is still under review.

In the area of detection using arrays to analyse the cDNA created from miRNA, companies such as Affymetrix, Exiqon, Life Technologies, Luminex (bead-based) and Agilent have adapted their technologies based on immobilization of oligonucleotides on surfaces as capture. All these microarray-based technologies have been employed for many years in the area of gene-expression profiling to study mRNA expression without being able to deliver robust products in the area of clinical diagnosis. It is expected that the lessons learned from mRNA platforms will result in fewer problems associated with reproducibility among novel array-based nucleic acid technologies in the future. However the big issue of all of these technologies will remain the same than before as they are based just in a single step, hybridization between targets and capture probes, to give an answer. This means that their current miRNA products do not offer any further innovation if compared with their microarray products for mRNA profiling, expecting then similar problems. This lack of innovation will be a blocking point for them to deliver robust and cost-efficient products for diagnoses and prognosis of cancer based on miRNA analysis. However, this innovation is essential to facilitate Stratified Medicine Programmes which goal is to help doctors to decide the most suitable treatment for patients based on genetic tests, being miRNA profiling tests clearly a need. As Lowe stated very recently "Further advances in the technology of miRNA profiling could help to revolutionize molecular pathology".¹⁴

The drawbacks of the current technologies provide a real opportunity for technologies such as the presented in this proposal which are capable of developing and introducing innovative solutions. The presented technology is able to identify nucleic acid sequences with very high specificity due to his unique chemical-based approach for nucleic acid "reading" which does not allow "mis-reading" and will provide solutions to obtain the needed sensitivity. Currently, the core technology has been validated by genotyping, with 100% accuracy, cystic fibrosis patients using MALDI-TOF as detection method. Herein we propose to develop new reagent sets to allow fluorescence-based detection to allow developing microarray formats and flow cytometry approaches in order to reduce the cost of assays and increase the number of applications. In recent years, we have created a game-changing technology for improved nucleic acid detection.¹⁵ This strategy applies dynamic chemistry to peptide nucleic acid (PNA) for the development of an entirely novel method for nucleic acid detection. This revolutionary approach was based on the hypothesis that Watson-Crick base pairing could be harnessed to template a dynamic reaction on a strand of PNA, by creating a nucleobase-free position on the PNA (a so-called 'blank' position) situated opposite to a nucleotide under interrogation on a complementary miRNA strand. A reversible reaction, between an aldehyde-modified nucleobases (**Figure 1 a**) and the free secondary amine at the blank position of the PNA probe would generate an iminium intermediate that can be reduced and analyzed (**Figure 1 b**). In the presence of all four aldehyde modified nucleobases

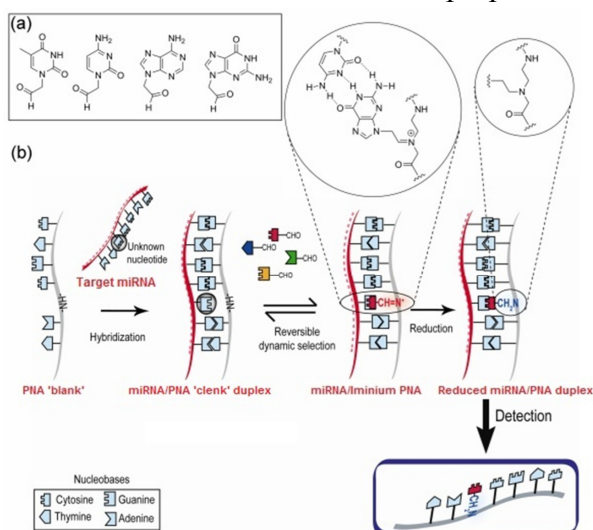


Figure 1(a) Aldehyde modified nucleobases. These molecules could conceivably be labelled with fluorophores to facilitate fluorescent detection. **(b)** Sequence specific miRNA-templated reductive amination on a PNA backbone possessing a 'blank' position constituting a free secondary amine.

based detection to allow developing microarray formats and flow cytometry approaches in order to reduce the cost of assays and increase the number of applications. In recent years, we have created a game-changing technology for improved nucleic acid detection.¹⁵ This strategy applies dynamic chemistry to peptide nucleic acid (PNA) for the development of an entirely novel method for nucleic acid detection. This revolutionary approach was based on the hypothesis that Watson-Crick base pairing could be harnessed to template a dynamic reaction on a strand of PNA, by creating a nucleobase-free position on the PNA (a so-called 'blank' position) situated opposite to a nucleotide under interrogation on a complementary miRNA strand. A reversible reaction, between an aldehyde-modified nucleobases (**Figure 1 a**) and the free secondary amine at the blank position of the PNA probe would generate an iminium intermediate that can be reduced and analyzed (**Figure 1 b**). In the presence of all four aldehyde modified nucleobases

(i.e. adenine, thymine guanine and cytosine derivatives) the nucleic acid template would stabilize the iminium species with the correct hydrogen-bonding motif (obeying Watson-Crick base-pairing), thereby amplifying it over the other possible iminium products. This would be reflected in the final product distribution after reduction to the corresponding amine, and would provide a means of nucleic acid discrimination for miRNA analysis.¹⁵⁻¹⁷ The specificity of this novel chemical-based method comes from two steps:

1. Selective hybridisation between high affinity molecular probes and target nucleic acid such as miRNA. Therefore the right oligomer length and design will tackle any homology issue.

2. Proof-reading step - selective dynamic incorporation of a specific nucleobase. The dynamic chemical reaction can ONLY take place when the target nucleic acid forms a perfect duplex with a designed PNA probe. This means that a signal is ONLY possible if the target nucleic acid is present and therefore it does avoid false-positives.

Other advantages of the proposed methodology are:

- (i). Easiness and speed of testing.
- (ii). Avoidance of false-positives, ensured by high affinity recognition of the specific nucleobase into PNA probe blank position
- (iii). Flexibility to design “personalized” and low cost multiplex assays for targets detection in single individuals
- (iv). Adaptation of the target detection technology to commonly used analysis platforms such as microchip readers and flow cytometers

The successful completion of this proposal will lead to the design, set up and validation of multiplexed assays for profiling miRNAs which will be used in cancer diagnosis and prognosis. In particular, the project will lead to release of the following products:

- 1. A microarray assay for profiling of miRNA-21 and miRNA-10
- 2. A bead based toolkit for rapid detection of circulating levels of miRNA-21 and miRNA-10.

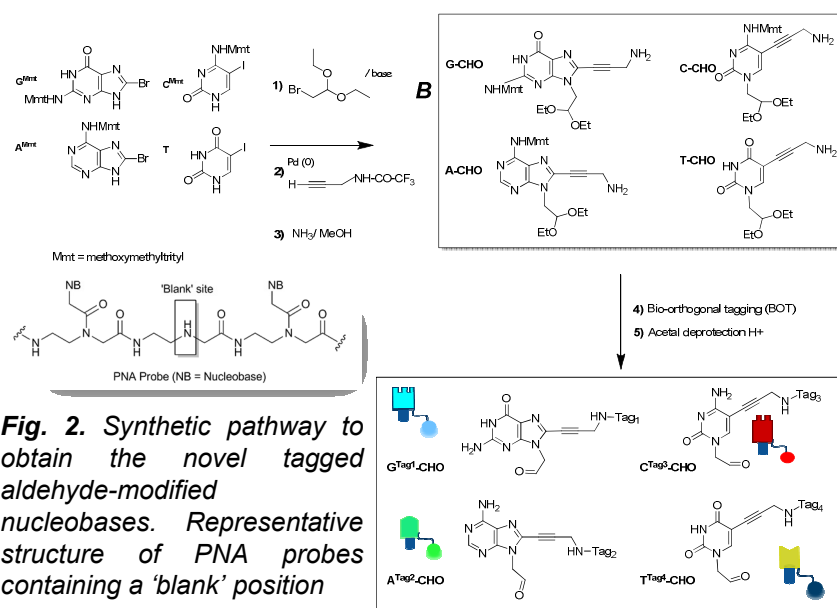
In summary, the special challenges that miRNA analysis presents lie outside the scope of molecular biology and instead involve the fields of chemical biology and biotechnology. However, the successful development of this project will require a multi-disciplinary effort involving not just chemical biologist but also biologist, clinicians and knowledge transfer experts to exploit the project results. *This joint effort will thus allow offering a robust and cost-effective miRNA profiling technology based on chemical-based reagents different to any other technology.*

B1.2 Appropriateness of research methodology and approach.

Aim 1 Design, synthesis and validation of chemical reagent sets for novel cost-effective miRNA profiling assays.

Previously, aldehyde-modified nucleobases without being tagged and abasic PNA probes have been reported by us.¹⁵⁻¹⁷ A key aim of this project is to create a second generation of reagents where modified nucleobases are labelled by fluorescence molecules or any other tag, such as biotin. The novel nucleobases will be prepared as shown in **Fig. 2**. We have experience in the preparation of these bases and do not envisage any significant problems with the synthesis. Furthermore, as shown in **Fig. 5** aldehyde-modified cytosine, tagged with a fluorescein molecule has been already prepared following the synthetic route depicted in **Fig. 2**. Briefly, after alkylation of the bromo and iodo-pyrimidine and purine derivatives, protected propargylamine will be introduced via classical Sonagashira coupling. Following deprotection of the exocyclic primary amine a coupling reaction will be carried out to introduce different tags. A final acetaldeprotection step will give rise to the target aldehydes. A range of tag will be explored such as fluorescein and biotin. This strategy is flexible enough to allow the rapid nucleobase-tagging as required by a given application, i.e. as

shown in Fig. xx all the nucleobases will be tagged by the same fluorophore. Other applications might require just one of them, the “Reader Base”, to be tagged and others the four of them tagged differently.



Probes containing a “blank” (abasic) position will be designed and synthesised so that they can hybridize perfectly to their specific miRNA targets recognised as oncomiR, i.e. miR-21 and miR-10.¹⁸⁻²⁰ Design will be done using publically available data bases such as NCBI and computational tools such as BLAST will be used to avoid homology

issues. Initially, probes targeting miR-21 and miR-10 will be prepared and validated using synthetic RNA strands and miRNA extracted from cell lysates (primary glioblastoma cells are known to have high levels of miR-21 and miR-10 – a collaboration with the group of Dr Angel Concha will allow the researcher to get access to these material which are kept in the Andalusian Biobank (Regional Ministry of Health – Consejería de Salud de la Junta de Andalucía), in particular in the Node Hospitalario Virgen de las Nieves where Dr. Concha Lopez acts as Director. Usages of human samples belonging to the Andalusian Biobank for biomedical research are strictly regulated by the regional and national legislation :

- Real Decreto de Biobancos 1716/2011 - published in BOE (Boletín Oficial del Estado) on 18th November 2011.
- Ley de Investigación Biomédica 14/2007, published in BOE on 3rd of July 2007.

A collaboration agreement between Dr. Concha and the PI has been put in place (please see Annex 4) and the project has already been submitted to the local ethical committee for its ethical approval (please see Annex 5).

The first part of the proposal is focussed on the profiling of oncomiRs, initially miR-21 and miR-10, which then will be used as a starting point for any other probe to be developed in aims 2 and 3. This methodology is well validated and has been successfully used for genotyping cystic fibrosis patients using cheek swabs as DNA source and MALDI-ToF as reading tool¹⁷ and, therefore, validation of the novel reagent toolkits will be done using MALDI-ToF as previously described.

Aim 2. miRNA-microsphere flow cytometry system for an *in vivo* miR-21 monitoring

The aim of this strategy is to apply, exploit and develop a microsphere miRNA-capture system that will allow the monitoring of miRNAs obtaining from both *in vitro* and *in vivo* sources using the dynamic chemical approach. The synergistic combination of this novel device and the chemical-based profiling approach will revolutionise the way of profiling miRNAs as it will allow direct extractions and detection of miRNAs from whole tissue.

Microspheres with a huge variety of compositions and sizes are currently used in biological processes ranging from controlled drug delivery devices to tools for quantification and analysis.²¹⁻²³ For example Dyna microspheres are used for a variety of magnetic based separations, usually *via* antibody based selection²² and large microspheres (often biodegradable) have been used extensively for drug delivery.²³ Over the past 9 years Dr Rosario M. Sanchez-Martin (RMS-M) has developed

chemistry methodology²⁴⁻²⁷ for the preparation of a range of mono-disperse, cross-linked, functionalised microspheres in a variety of user defined sizes (200 nm-5 µm), which show both chemical and biological compatibility.

This research intends to develop improved methods of miRNAs detection by using high-affinity molecular probes suitable for the chemical based detection of nucleic acids supported on surface beads - including magnetic beads - to obtain a simpler and faster protocols as well as higher multiplexing capability.

The researched will use PNA conjugated magnetic microspheres as devices for the precise capture of miRNAs directly from cell lysate, blood or **urine from the Andalusian Biobank (see page 6 for details)**. RMS-M has already demonstrated the versatility of the use of microspheres for covalent conjugation of a wide range of biocargoes (see reference 28 for a overview of oligonucleotide conjugation.²⁴⁻²⁹ These strategies will be adapted to allow conjugation with high affinity molecular probes complementary to significant miRNAs.

Once this has been optimised the capture of miRNAs directly from body fluids (blood, urine) or cell lysate **from the Andalusian Biobank (see page 6 for details)** will be explored as shown in Fig. 4. Successively, double helixes (PNAs/miRNAs) from whole the blood bound will be separated using magnetic separation. The PNA probes contain the 'blank' (**Figure 1**) position situated opposite to a nucleobase on a complementary miRNA strand. A reversible reaction, between an aldehyde-fluorescein nucleobase (**Figure 3**) and the free secondary amine at the blank position of the PNA probe would generate an incorporation of a fluorescein nucleobase-conjugated. Finally, the so now fluorescein-labelled microspheres will be sorted by flow cytometry. Also here, as mentioned above, quantification will be made by using microspheres-PNA probes which will identify "housekeeping" microRNAs or by spiking reactions with known nucleic acids.

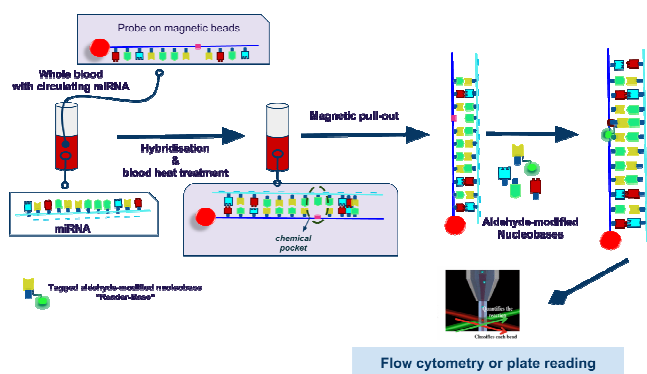


Figure 4 Components of the miRNA-microsphere flow cytometry system for an *in vivo* miR-21 monitoring. Magnetic bead - PNA probes with a 'blank' position that hybridizes to the target miR-21 (red). Inclusion of an aldehyde-modified nucleobase (with fluorescein) to the 'blank' position will make microsphere sortable by flow cytometry.

The multiplexing capability of bead-based approaches come from the ability of optical-encoding of beads. Once

validated these novel devices the technology will be developed further to increase its multiplexity capability and sensitivity by using commercial available encoded-particles.

- (i) Colour-coded beads (Luminex) will be conjugated to high affinity molecular probes, being each of these probes associated to one specific colour-coded bead. In this way up to 100 different miRNAs could be analysed per sample.
- (ii) Bar-coded gel microparticles (Firefly Bioworks) will also be used to conjugate probes in a similar fashion that in (i) and with higher multiplexy and lower detection limit than (i).

*This method will allow rapid detection of miRNAs (in one single read directly from biological sources such as blood, urine, cell lysate) **from the Andalusian Biobank (see page 6 for details)** without a pre-isolation of microRNAs, and be of particular use in monitoring in vivo levels of circulating miRNAs such as miR-21. Combining the high specificity of the chemical-based approach (which avoids false-positives) with the high sensitivity and multiplexy capability of particle-based technologies will provide a tool with strong predictive, diagnostic and prognostic*

potential due to their robustness (lack of false-positives and reducing number of sample manipulation).

B1.2.3 Novel miOncochip for oncomiRs detection

The final aim of this project is to develop an efficient microarray-based detection platform, so-called “miOncochip”, for detection of miRNAs associated to cancer. The concept introduced here will be a combination of the standard microarray approaches in association with the chemical-based technology for nucleic acid testing to allow detection and quantification of miRNAs with improved speed of analysis and absence of false positives.

The core principle behind microarrays is the hybridization between two DNA strands, the property of complementary nucleic acid sequences to specifically pair with each other by forming hydrogen bonds between complementary nucleotide base pairs. After washing-off of non-specific bonding sequences, only strongly paired strands will remain hybridized. So fluorescently labelled target sequences that bind to a probe sequence generate a signal.³⁰ However, these technologies require complex and expensive processes for targeted nucleic acid labelling, especially in the case of miRNA, and a very fine temperature control to avoid mis-hybridization which give rise to false positives as just a single hybridization step is responsible of the signal. This is particularly difficult when high homology sequences have to be analysed. It would be exceptionally useful to be able to miniaturize oncomiRs detection in devices known as biochips, including microarrays and lab-on-a-chip (LoC) microfluidic devices, such as the In-Check Platform from ST Microelectronics, without the risk of false positives and being able to use label-free miRNAs. The proposed project offer such solution.

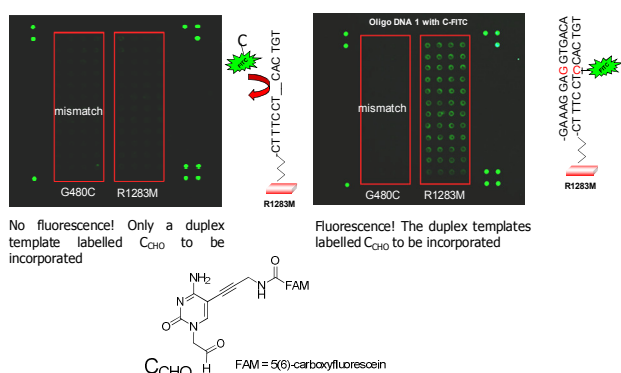


Figure 5 Recent work shows the core technology working on a microarray format to identify allelic variants.

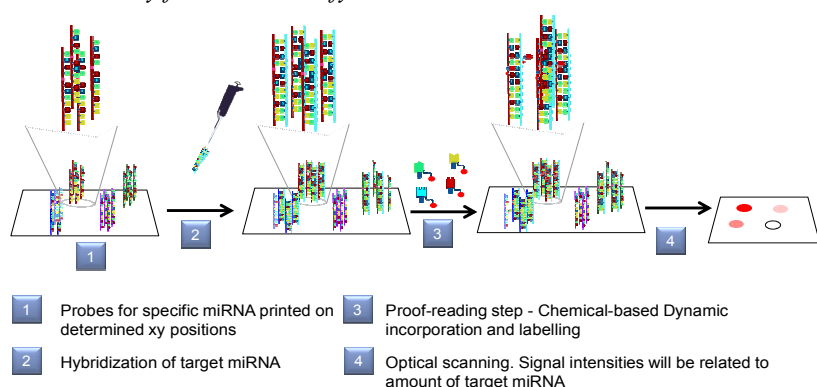


Figure 6 Components of the “miOncochip” probe design. (1) Probes printed on a microarray format at a ‘blank’ position that hybridizes to target miRNA. Probes will be anchored to the glass slide surface by a PEG linker. (2) Label-free miRNAs will be hybridized on the array (3) Proof-reading step - dynamic incorporation of a fluorescently-labelled aldehyde-modified nucleobases (it can be any of the four) to the ‘blank’ position will only happen when a duplex is formed. When a miRNA is not present then a signal will not be generated (4) This fluorescently-labelled nucleobase incorporated in the probe-target interaction region will be followed by optical scanning.

Recent data produced by the candidate demonstrated the capability of the core technology to identify allelic variants linked to cystic fibrosis on microarrays using a fluorescently-labelled aldehyde-modified cytosine. **Figure 5** (data unpublished).

In the first stage PNA probes will be printed on different both contact arrayer and inkjet printers (an area in which the candidate have a vast experience). Printing will be done in University of Granada at Fundacion Medina and also in collaboration with Prof. Bradley, the co-inventor of the core technology at the University of Edinburgh. Once optimised microarrays will be manufactured either by STMicronics or Panagene (Panagene have already supplied the candidate with PNA probes containing ‘blank’ positions and they do not expect any problem to manufacture these arrays).

Microarrays will be initially interrogated using synthetic. Once this has been optimised, the miR-21 and miR-10 profiling will be investigated. Samples will be

provided either from *in vitro* (miRNAs isolated from cell culture) and *in vivo* (miRNAs isolated from transgenic mice overexpress miR-21 blood samples). Animals will not be bred by the PI during the project. Blood samples from transgenic mice will be bought from The Jackson Laboratory, an US independent, non-profit organization focusing on mammalian genetics research to advance human health. The Jackson Laboratory is fully accredited by the Association for Assessment and Accreditation of Laboratory Animal Care International (AAALAC International) and also have have an Animal Welfare Assurance with the Office of Laboratory Animal Welfare (OLAW). They offers tissues and biospecimen such as blood samples from its Repository-Live which contains mice useful in miRNA biology. miRNA analysis of those biospecimens using the new tools and platforms developed during the project will be carried out at the unit of cell culture experimentation of the Center of Scientific Instrumentation of the University of Granada that it is equipped with CLASS II biosafety cabinets. The project has already been submitted to the local ethical committee for its ethical approval (please see Annex 5). Dr. Medina, an expert in miRNA animal models,⁶ is currently in Granada with a Ramon y Cajal Researcher contract as the candidate has and he has already agreed to assist the candidate in this area ~~an approach to collaborate with him~~). Molecular probes to profile miR-21 and miR-10 will be immobilised, fluorescently-labelled aldehyde-modified nucleobase(i.e. guanine) will be incorporated and following scanning quantified (reactions will be spiked to have internal controls for quantification). This approach could be expanded to other targets

After proof of concept experiments, the method will be evaluates and validated on a pool of different miRNAs¹⁹⁻²⁰ and by labelling each nucleobase with different colors it could also be suitable to analyse miRNAswhcih differ in a single base as in the case of let-7 miRNAfamily(Fig. 6 shows a cartoon representing a 4-color miOncochip).

A successful execution of this project will produce an ultra-sensitive and specific miOncochip, which would be able to discriminate between very similar sequences thanks to its unique internal proof reading system. This technology will be suitable for biochip-based one-step point-of-care-diagnostic test to analyse samples of serum, or blood, for the presence of miRNAs markers.

B1.3 Originality and innovative nature of the project, and relationship to the 'state of the art' of research in the field

The core technology, invented by the candidate and Prof. Bradley (PCT/GB2008/003185) allows a direct read-out of short nucleotides sequences without need for additional probe/target recognition, PCR amplification or labelling of the targeted nucleic acid, making it ideal for miRNA profiling. The technology has several important features including the ability to eliminate false positive results (as there is a thermodynamic process which control the nucleobase recognition and therefore avoiding the mis-incorporation), which are a common feature of other detection technologies (eg PCR and arrays). The technology shows both high specificity and sensitivity, and is a highly novel and innovative technique. To date the technology has been validated for cystic fibrosis genotyping using mass spectrometry, MALDI-ToF, to detect the assay signal and outcome. A potential limitation to the widespread adoption is that mass spectrometry is required in order to read the signal or assay output. Therefore, this project propose a key innovative step to apply fluorescence detection methods in order to make the assay cheap and easy to perform without the use of expensive and specialist equipment such as MALDI-TOF. This innovation will greatly facilitate the integration of the technology within state of the art platforms providing them with an unique way of solving traditional problems associated to microarray an bead based detection systems. The new platforms will be the very first one which are not based on just hybridization process but also on a second step which will eliminate the risk of false positives.

The originality and innovation of the project will produce an impact on three different areas:

- (i). *Scientific and Translational impact* The ability to detect multiple and key molecular targets associated to pathologic progression is essential to this aim. Another primary scientific goal will be the validation of a circulating miRNA signature for diagnosis of cancer.
- (ii). *Technologic impact* The release of the chemical toolkits proposed here will pave the way for the birth of “research-lab independent” technology for performing nucleic acid-based diagnosis in large patients cohorts and
- (iii). *Diagnostics and Clinical impact* The use enzyme-free nucleic acid detection methods, such as that proposed here, is fated to radically change the field of molecular diagnostics.

B1.4 Timeliness and relevance of the project

The ever increasing relevance of miRNA and their linkage to personalised and stratified medicine will mean that this technology will become a key partner of drug discovery companies and national health services. Programs launched very recently such as the Stratified Medicine Program by Cancer Research UK which brings together AstraZeneca, Pfizer and the UK government's Technology Strategy Board aim to establish the foundations for national services that will ensure standardised, high quality, cost-effective genetic testing of tumours is available for people with cancer. As currently there are not gold standard technologies to do miRNA analysis there is a clear window of opportunity for this technology to be established itself as one of the standards technologies which will be used for cancer stratification and tailored cancer drug treatments, helping the EU to be a reference in the new era of medicine. The innovation presented here could be the one which could help to revolutionise the field of molecular pathology linked to cancer as Lowe has just predicted.

B2 RESEARCHER

B2.1 Research career potential

The development of this project will unquestionably help to the reintegration of the researcher as it will offer him the opportunity to settle down as an independent researcher in the new place of work, supervising students and starting a research line to continue with his scientific career. The development of this project will allow the researcher to adopt a privileged position in a state-of-the-art field of the current research and be competitive and productive within the scientific community. The researcher will be willing to collaborate within the world scientific community and expanding his current scientific network from the UK to Spain. Also the development of this project will allow the researcher to established local relationships with research group lead by Dr. Angel Concha Lopez's (who is Chief of the Pathology Services and Director of Biobank of cells, tissues and tumours of the Hospital Virgen de las Nieves de Granada), with Dr. Jose Luis Garcia Perez, who recently received an awarded by the Howard Hughes Medical Institute with an Early Career Award which is awarded to people who are expected be the scientific leaders in their countries selected and who works at the Pfizer - Universidad de Granada - Junta de Andalucía Centre for Genomics and Oncological Research (GENYO) in Granada. On top of that it will also allow the candidate to keep a close working relationship with Dr. Rosario M. Sánchez-Martín who is a world leader in the field of nanoparticles and their biological applications and Lecturer in the School of Pharmacy at the University of Granada. These collaborations are an excellent chance for the researcher and the investigators involved in the project to acquire high quality training, as well as it will open a wide range of possibilities for future research implementation. The integration in the Department of Medicinal and Organic Chemistry at the University of Granada will allow the candidate to acquire new knowledge and more experience in different fields, being a clear opportunity to perform basic research with medicinal chemistry goals that will be carried out to move forward in his career to become and independent researcher. This proposal combines his previous research experience with those of the technologies available at the Department of Medicinal and Organic Chemistry and other research centres at the University of Granada and thus is ideally suited for the consolidation of his independent research career. Such a multidisciplinary project will utilise his core skills, allowing him to further develop his interests in cell and tissues based assays, while enabling him to collaborate directly with biologist and clinicians Dr. Díaz-Mochón will received training in specific fields such as chemical strategies for PNA conjugation to magnetic microspheres and them the use of flow cytometry, handling of biological samples such as preparation of cells lysates, i.e from glyoblastoma cell lines, for miRNA extraction and training in the last techniques within the genomic analysis area, such as deep DNA and RNA sequencing. To this aim, the candidate will work closely with Dr. Rosario María Sánchez Martín (expert in chemical strategies in nanotechnology from the University of Granada), Dr. Angel Concha (Chief of the Pathology Services and Director of Biobank of cells, tissues and tumours of the Hospital Virgen de las Nieves de Granada, Andalusian Health Service) and Dr. Jose Luis García Pérez (Head of the reprogramming and retrotransposition genomic unit of the Genyo) respectively.

To achieve this aim the innovative nature of the current proposal project plays a main role. The research proposed represents a new topic in the medicinal chemistry field that will allow the candidate to become an independent scientist and contribute to enhance EU scientific excellence. The applicant has received training in Chemical Biology as outlined above so, on my return, he will be able to bring to the University of Granada this new expertise with respect to the design and preparation of novel chemical-based tools for biological and medical applications. This project will give him the chance to initiate independent research and integrate the training from the previous fellowship with my own research ideas. Thus the research project presented in this proposal has been developed coherently with regards to my research experience. Regarding the professional career of the candidate, the host institution efforts will be focus on aims to develop the skills of the researcher to become an independent team leader upon the completion of the reintegration period. Currently, the researcher has a five year contract as Ramon y Cajal Researcher (a competitive contract within a Spanish National Programme for Recruitment and Incorporation of Human Resources) at the University of Granada, starting June 2011through May 2016. During this time, the candidate will be able to apply for an upgrade that would result in a permanent contract. From there, the applicant could apply for another upgrade to Lecturer position. All of these successive positions involve the development of research and teaching and from all of them the applicant can run a research group and be an independent researcher.

B2.2 Research and technological quality of previous research

Between my major accomplishments, three can be distinguished:

1.-Development of High Throughput Screening Platforms for both enzymatic and cell-based assays. This includes PNA-encoding libraries and their interfacing with DNA microarrays for decoding and cell-based polymer microarrays. The PNA-encoding technology has been demonstrated to be successful to find out new substrates for proteases (thanks to this platform we have presented the first specific substrate for chymopapain) and kinases (an Oncochip based on this technology is currently been developed within the Dutch CTMM and partners). Furthermore and very recently we have presented very exciting results, where by adapting this technology to be used with cells, we have found peptides which act as cell specific receptor ligands and as tumour-homing peptides, allowing the targeting of tumoral cell lines with peptides discovered with this platform. When I started my post-doctoral period with Prof. Bradley this was just a concept and to date PhD students, post-doctoral researches and projects worth over 4million EUR have been generated at the back of this project. I have also key to develop polymer microarrays for cell-based assays. With the know-how and equipment created by the previous project we created polymer microarrays which allow discovering new biomaterials for cell lines, primary and stem cells. One of the major achievements in this area has been the discovery of a new material which allow growing of hepatocytes-like cells which facilitate the in-vitro study of toxicity of new drugs, a major world-wide issue as hepatocytes can normally not be functional long enough for this kind of studies

2. Nucleic Acid Analysis by Dynamic Chemistry. During my last part of my research period at the University of Edinburgh I developed a new technology by which dynamic chemistry has been applied as an effective method of DNA and RNA analysis, demonstrating high base selectivity and the potential for enzyme-free SNP genotyping. The approach also raises the possibility of analyzing other sources of genetic variation and mutation, such as indels and abasic sites, in a manner simply not possible with current approaches. Moreover, it offers an approach to identify nucleobase mimics to expand the genetic alphabet. This game-changing research has already created a lot of opportunities in the academic world by the creation of PhD studentships, the clinical field with collaborations with world-class and the private sector with the creation of an spin-off company which task is commercialise this technology. This technology is flexible enough to be integrated within different platforms, as demonstrated in this proposal, and it will allow development of new tests in field so different such as cancer diagnosis and prognosis and infectious diseases and it will also allow development of low-cost tests for full-fill social needs such as tuberculosis and malaria detection products which require of low cost technologies to be implemented.

This has allowed the creation of high quality jobs which brings innovation to the EU

3. Knowledge Transfer. While I have spent a long period as academic researcher, firstly as post-doctoral researcher and after as PI of a project, one of my biggest achievements has been my training and skill development in the knowledge transfer area. Being PI of a project funded by Scottish Enterprise within their Proof of Concept Program allowed me to participate in different programs and meetings to learn concepts and realities related with this area. This was key for me to do the whole transition from lab bench to business creating a spin-off company, DestiNA Genomics, to exploit commercially one of my ideas. To do so, I have had to participate and define strategies for how to protect best intellectual properties, structuring them, from patents to know-how, in an adequate manner, how to negotiate with Universities licensing agreements to exploit a certain intellectual property piece, how to structure a company around a biotechnological licence, how to convince investor of the value of a biotech company, which most of the time have not revenue and therefore the valuation of the intangibles is the fine art which decide between success and failure. This transition allowed me to become Chief Scientific Officer of this biotech company which have managed to create eight jobs based in Scotland to date. I have also got one of my patent being taken by another spin-off working in the field of biomaterials.

B2.3 Independent thinking and leadership qualities

During my scientific career I have been moving through the ladder quite steadily. I arrived to the UK as scientific visitor and after 18 months I was offered a contract as research assistant to become staff member. Then 18 months latter I was offered to move to Edinburgh as Senior research assistant to the Bradley group,

a research group of over 30 members, where I acted as scientific manager. I run directly and supervising numerous research stands within the Bradley Group - pushing back the boundaries within the area of Chemical Biology and specifically High-Throughput Chemical Biology and the whole area of Chemical Microarrays (including polymer microarrays and PNA/DNA encoding arrays) and supervising five students directly in my own right. As such, with Prof. Bradley I have published together over 35 papers (many in the top chemical journals (e.g Angew Chemie) and patents (upon which we share co-inventorship) with some of the patents now licensed from the University of Edinburgh.

In 2007 I got the University approval to protect by filling a patent one of my ideas which latter gave me my own funding through the Scottish Enterprise Proof of Concept Program. I was able to raise more money (£65,000 to make it a total of £285,000) from the original budget. This milestone driven project produced excellent results scientifically, in terms of public awareness through my presence in mass media such as the BBC and obviously through the creation of a spin-off company. My role as CSO of DestiNA Genomics where I led a team of eight people in the scientific area, managed to secure a mix of private investment and public through receiving grants.

In parallel I am still Partner Coordinator of the CTMM project which is based in the PNA-encoding technology that I developed. In Edinburgh, there are two scientists working on this project under my partial supervision. Part of my responsibilities is to make sure that milestone set up for Edinburgh are met as well as coordinate this work with our Dutch partners.

I was part of the negotiating team from the School of Chemistry (Edinburgh) to get a collaboration project led by Prof.Tasker with Samsung Microelectronics (Korea). The School of Chemistry in Edinburgh has been pioneer in the use of the Chemical Manager Inventory, a web-based tool which allows the recording and tracking down of all the chemicals within the School and which have already brought spectacular results in terms of reducing the costs of chemical spending. Back in 2007 I formed part of the team lead by Dereck Burgess, which initiated the talks with SciQuest, the provider of the service. The Bradley group was the first research group, which tested the system before being expanded to the rest of the School under my coordination.

B2.4 Match between the fellow's profile and project.

During the last ten years, I have been working in the development of microarray platforms for high throughput screening of biological systems based on a PNA encoding strategy and also developing polymer microarrays for cell-based assays. This gives me the needed expertise to use and apply the biotechnological tools needed to carry out this project. A key element is that I have invented and developed the core technology used in this proposal. This core technology has been validated by an SNPs and indels from human cheek swabs that allow the diagnosis of severe diseases such as cystic fibrosis. The experimentation required to conduct the present project has been used widely by Dr. Diaz-Mochon during the last his career as a postdoctoral fellow and a research fellow at the Chemical Biology section of the School of Chemistry at the University of Edinburgh. I am also very familiar with the synthesis protocols needed to prepare these modified nucleobases and build the PNA probes described in this project, and have conducted extensive work in the development of microarray-based detection platforms as reflected in my CV. This project is novel, although is based in previously established techniques, and I do not anticipate many technical challenges that may prevent the consecution of the proposed aims. Besides the scientific skills that the current project requires I have also been involved in other areas of research which could facilitate certain tasks, such as the use of peptide and fluorescence (see CV in section B.2.5.). This experience and knowledge makes the researcher an excellent candidate to carry out the proposed research. The applicant has acquired the necessary experience and skills in a broad range of research areas to be able to develop this project.

B2.5 Curriculum Vitae

ACADEMIC ACHIEVEMENTS:

UNIVERSITY EDUCATION

1997-2001 PhD in Pharmacy (Medicinal Chemistry) Department of Medicinal and Organic Chemistry, University of Granada, Spain. Mark Obtained: *Cum Laude*.

1991-1996 Degree in Pharmacy. Mark Obtained: 3.1 (Scale 1-4) University of Granada, Spain. Equivalent to 1st Class Honours Degree.

RESEARCH EXPERIENCE

- **1995-1996:** Erasmus student at the Dipartimento di Chimica e Tecnologia del Farmaco (Department of Chemistry and Drug Technology), Università degli Studi di Perugia. Perugia, Italy, under the supervision of Prof. Pellichari.
- **1997- 2001:** PhD student in the Department of Medicinal and Organic Chemistry, Faculty of Pharmacy, University of Granada, under the direction of Prof. Espinosa Ubeda, Head of Department.
- **2000:** visitor research student at the Dipartimento di Scienza Farmaceutica, Università degli Studi di Ferrara. Ferrara, Italy, under the supervision of Prof. Baraldi.
- **June 2001- January 2002:** postdoctoral fellow in the Department of Medicinal and Organic Chemistry, Faculty of Pharmacy, University of Granada, under the direction of Prof. Espinosa Ubeda, head of department.
- **February 2002- June 2003:** Post-doctoral research visitor at the School of Chemistry, University of Southampton, under the supervision of Prof. Mark Bradley.
- **July 2003- January 2005:** Post-doctoral research assistant at the School of Chemistry, University of Southampton, under the supervision of Prof. Mark Bradley.
- **February 2005-June 2008:** Post-doctoral research assistant (PDRA) at the Chemical Biology section, School of Chemistry, University of Edinburgh, under the supervision of Prof. Mark Bradley.
- **July 2008-October 2010:** Researcher Fellow of the Chemical Biology section, School of Chemistry, University of Edinburgh. Line Manager: Prof. David Leigh; Mentor: Prof. Mark Bradley.
- **November 2010-May 2011:** Chief Scientific Officer, DESTINA Genomics Ltd. Spin-off company of the University of Edinburgh.
- **June 2011-May 2016:** Ramón y Cajal Researcher, Department of Medicinal and Organic Chemistry, Faculty of Pharmacy, University of Granada.

FELLOWSHIPS AND OTHER AWARDS/ (External Recognition/steem)

- **2011** Honorary Fellow of the University of Edinburgh
- **2009** Promotion Award to grade 8 in the 2009 Academic Promotions & Gradings, University of Edinburgh.
- **2002** Outstanding Thesis Award 2000/2001 (First Prize) by the University of Granada, Spain. Thesis Title: *Novel specific antitumoral compounds against ras oncogen transformed cells targeting Choline kinase.*
- **2000** Financial backing for a short research stay abroad (Prof. Baraldi, Dipartimento di Scienza Farmaceutica, Università degli Studi di Ferrara. Ferrara, Italy). From 01/05/2000 to 31/07/2000.
- **1997-2001** Regional Government Grant for PhD studies (Junta de Andalucía, Spain). From 01/09/1997 to 31/06/2001.

Group and Associations

- Member of the Royal Society of Chemistry (RSC), United Kingdom.
- Member of the International School of Advanced BioMedicine and Bioinformatics, University of Catania, Italy.

LIST OF RESEARCH PUBLICATIONS (ONLY IN LAST THREE YEARS)

Author of 36 peer-reviewed articles: 90% of them are classified as Q1 in the Journal Citation Reports (JCR)2010, SCI and 3 of them as corresponding author. H index =14

Research publications last three years are:

Articles in scientific journals

1. **Diaz-Mochon, J.J.**; Planth, S.; Bradley, M. From 10,000 to 1: Selective synthesis and enzymatic evaluation of fluorescence resonance energy transfer peptides as specific substrates for chymopapain, *Analytical Biochemistry*, **2009**, 384, 101-105 (5 – Year Impact Factor 3.343 JCR).
2. Pernagallo, S.; **Diaz-Mochon, J. J.**; Bradley, M. A cooperative polymer-DNA microarray approach to biomaterial investigation, *Lab on a Chip*, **2009**, 9, 397-403 (5 – Year Impact Factor 6.928 JCR)
3. Tourniaire, G.; **Diaz-Mochon, J. J.**; Bradley, M. Fingerprinting Polymer Microarrays, *Combinatorial Chemistry & High Throughput Screening*. **2009**, 12, 690-696. (5 – Year Impact Factor 2.190).
4. Bowler, F.; **Diaz-Mochon, J. J.***; Swift, M.D.; Bradley, M., M. DNA Analysis by Dynamic Chemistry, *Angewandte Chemie International Edition*. **2010**, 49, 1809-1812 (5 – Year Impact Factor 12.050) (**corresponding author**).

5. Svensen, N.; **Diaz-Mochon, J. J.**; Bradley, M. Decoding a PNA encoded Peptide Library by PCR: the discovery of new cell surface receptor ligands, *Chemistry & Biology.*, **2011**, 18, 1284-1289. (5-Year Impact Factor 5.976).
6. Svensen, N.; **Diaz-Mochon, J. J.**; Bradley, M. Microarray Generation of Thousand-Member Oligonucleotide Libraries, *PLoS ONE*, **2011**, 6(9), e24906 (5-Year Impact Factor 4.610)
7. Bowler, F.R.; Reid, P. A.; Boyd, C. A.; **Diaz-Mochon, J. J.***; Bradley, M. Dynamic Chemistry For Enzyme-Free Allele Discrimination In Genotyping By Maldi-Tof Mass Spectrometry, *Analytical Methods*, **2011**, 3(7), 1656-1663 (5-Year Impact Factor 10.036) (**corresponding author**).
8. Svensen, N.; **Diaz-Mochon, J. J.**; Bradley, M. Encoded Peptide Libraries And The Discovery Of New Cell Binding Ligands, *Chemical Communications*, **2011**, 47(27), 7638-7640 (5-Year Impact Factor 5.582).
9. Svensen, N.; **Diaz-Mochon, J. J.**; Dhaliwal, K.; Planonh, S.; Dewar, M.; Armstrong, J.D.; Bradley, M. Screening Of a Combinatorial Homing Peptide Library For Selective Cellular Delivery, *Angewandte Chemie-International Edition*, **2011**, 50(27), 6133-6136 (5-Year Impact Factor 12.050).
10. Hay, D. C; Pernagallo, S.; Díaz-Mochón, J. J.; Bradley, M. Unbiased Screening Of Polymer Libraries To Define Novel Substrates For Functional Hepatocytes With Inducible Drug Metabolism, *Stem Cell Research*, **2011**, 6(2), 92-102 (5-Year Impact Factor 4.107).
11. Unciti-Broceta, A.; **Díaz-Mochón, J. J.**; Sanchez-Martin, R.; Bradley, M. The Use of Solid Supports to Generate Nucleic Acid Carriers, *Accounts of Chemical Research*, **2012**, in press (5-Year Impact Factor 20.330).

Book chapters

1. Pernagallo, S.; **Diaz-Mochon, J. J.***; Polymer microarray for High content cellular Screening; Book title: Cell-Based Microarrays. Collection: Methods and Protocols (Methods in Molecular Biology), Palmer, E. Editor, HUMANA PRESS, 2011, Volumen 706, 171-180. (**Corresponding author**).

Patents

1. Inventors (p. o. de firma): Bradley, M. and **Diaz-Mochon, J.J.**

Title: Nucleobase Characterisation. Application Form number: PCT/GB2008/003185. Priority Country: United Kingdom. Priority date: 17/09/2007.

Holder Entity: The University Court of the University of Edinburgh

OTHER COUNTRIES WHICH THE PATENT HAS BEEN EXTENDED TO: International patent.

COMPANIES THAT ARE LICENSING IT:: Destina Genomics Ltd.

2. Inventors (p. o. de firma): Hay, D. Pernagallo, S. **Diaz-Mochon, J.J.** Bradley, M and Iredale, J.

Title: Polymer for growing cells. Application Form number: PCT/GB2010/000523 Priority Country: United Kingdom. Priority date: 20/03/2009.

Holder Entity: The University Court of the University of Edinburgh

OTHER COUNTRIES WHICH THE PATENT HAS BEEN EXTENDED TO: International patent.

COMPANIES THAT ARE LICENSING IT: FibromEd Ltd.

LIST OF PARTICIPATION IN RESEARH PROJECTS

I have been part of the research team developing and working on the following research grants:

1. Research project title: PCR free nucleic acid detection and identification as a revolutionary molecular diagnostic tool. Financial entity: Scottish Enterprise – United Kingdom. Participant entities: DestiNA Genomics ltd. Length: from 01/01/2011 to: 31/12/2011. Funding:100.000 GBP. Principal Investigator: **Juan J. Díaz Mochón**
2. Research project title: Novel error free test for detecting norovirus in food. Financial entity: UK Technology Strategy Board– United Kingdom. Participant entities: DestiNA Genomics Ltd y Glycomar Ltd. Length: from 01/05/2011 to: 31/07/2011. Funding:24.575 GBP. Principal Investigator: Charlie Bavington.

3. Research project title: Novel Companion Biomarker Assay for Glioma Cancer Drug Development to Overcome Current lack of in-vivo Test Success. Financial entity: UK Technology Strategy Board– United Kingdom. Participant entities: DestiNA Genomics Ltd y University of Edinburgh. Length: from 01/05/2011 to: 31/07/2011. Funding: 33.000 GBP. Principal Investigator: **Juan José Díaz Mochón**.

4. Research project title: Chemical based DNA sequencing and SNP analysis. Financial entity: UK Scottish Enterprise – United Kingdom. Participant entities: DestiNA Genomics Ltd y University of Edinburgh. Length: from 22/07/2008 to: 21/07/2010. Funding: 285.000 GBP. Principal Investigator: **Juan José Díaz Mochón**.

5. Research project title: : CTMM BioCHIP: The Oncokinase Chip (WP2). Financial entity: Center for Translational Molecular Medicine (CTMM) – Dutch. Participant entities: University of Edinburgh. Length: from 01/09/2008 to: 31/08/2013. Funding: 538.778 euros. Principal Investigator: Joop Van Helvoort

6. Research project title: : High-Throughput Chemical Biology for Efficient Cellular Delivery Financial entity: Engineering and Physical Sciences Research Council (EPSRC)- United Kingdom. Participant entities: University of Edinburgh. Length: from 01/09/2005 to: 21/07/2008. Funding: 418.096 GBP. Principal Investigator: Prof. Mark Bradley

7. Research project title: : Interfacing DNA arrays and Combinatorial Libraries for High-Content Enzymatic Screening. Financial entity: Biotechnology and Biological Sciences Research Council (BBSRC) - United Kingdom and the University of Southampton. Participant entities: University of Edinburgh. Length: from 01/07/2003 to: 31/08/2005. Funding: 513.485 GBP. Principal Investigator: Prof. Mark Bradley.

8. Research project title: : Development of new COX-2 inhibitors. Financial entity: LABORATORIOS FAES FARMA S.A. (BILBAO) REF: F1825- Spain. Participant entities: LABORATORIOS FAES FARMA S.A. and the University of Granada. Length: from 01/07/2001 to: 31/01/2002. Principal Investigator: Prof. Antonio Espinosa Úbeda.

9. Research project title: : Design of compounds with antitumoral activity that interfere with the signal pathway induced by oncogens (Diseño De Compuestos Con Actividad Antitumoral Que Interfieran Con El Sistema De Transmisión De Señales Inducida Por Oncogenes: Diseño Y Síntesis De Nuevos Compuestos). Financial entity: Ministerio de Educación y Ciencia - Spain. Participant entities: University of Granada. Length: from 01/06/1998 to: 01/06/2001. Funding: 51146.13 euros. Principal Investigator: Prof. Antonio Espinosa Úbeda.

PARTICIPATIONS IN CONFERENCES (ONLY IN LAST THREE YEARS)

Author of 25 conferences contributions, including 5 invited international talks and a total of 14 oral communications at international conferences and 11 posters presented at international conferences. The participations in conferences in the three previous years are:

1. **Díaz-Mochón, J.J.** ORAL COMMUNICATION. PCR-free Nucleic Acid Detection. *6th Course of The International School of Advanced Molecular and Systems BioMedicine: Molecular Systems BioMedicine And Complex Pathological Phenotypes*, Catania (Italy), **2011**.

2. **Díaz-Mochón, J.J.** and Ilyine, H. ORAL COMMUNICATION. Route to Markets: Commercialising a chemical based nucleic acid detection platform. *Business Models: Adapting to Changes 8th Young Company Finance*, Stirling (United Kingdom), **2010**.

3. **Díaz-Mochón, J.J.** and Ilyine, H. ORAL COMMUNICATION. Investing on a Revolutionary Nucleic Acid Detection Technology *Connect Scotland Springboard & Investment Conference*, Edinburgh (United Kingdom), **2010**.

4. **Díaz-Mochón, J.J.** ORAL COMMUNICATION. DNA Analysis by Dynamic Chemistry *16th RSC Perkin Meeting - 38th Scottish Regional Meeting of the Organic Division of the Royal Society of Chemistry* Dundee (United Kingdom), **2009**.

5. D C Hay, S Pernagallo, **J. J Díaz Mochón**, J Schrader, J Fletcher, D Dalgetty, C Medine, J B Black, A I Thompson, Z Hannoun, P N Newsome, S J Forbes, J A Ross, M Bradley, J P Iredale POSTER. A simple polyurethane matrix promotes hepatic endoderm viability and inducible drug metabolism: implication for drug toxicology testing and the design of liver support devices *Annual Conference of the Asia-Pacific-Association-for-the-Study-of-the-Liver* Beijing (China), **2009**.

6. D C Hay, S Pernagallo, **J. J Díaz Mochón**, J Fletcher, J B Black, A I Thompson, Z Hannoun, P N Newsome, S J Forbes, J A Ross, M Bradley, J P Iredale POSTER. Polymer library screening identified an extracellular matrix that promotes and stabilises human embryonic stem cell-derived hepatocyte function. *44th Annual Meeting of the European Association for the Study of the Liver (EASL)* Copenhagen (Denmark) **2009**.

7. J. J Díaz Mochón ORAL COMMUNICATION. Diagnostic Application of PNA Technology. *5th International School of Advanced Biomedicine and Bioinformatics 5th Course - Complex Systems BioMedicine: Molecules, Signals, Networks, Diseases* Catania (Italy) **2009**.

8. J. J Díaz Mochón ORAL COMMUNICATION. Interfacing Whole Cells with PNA-Encoded Peptide Libraries. *5th International School of Advanced Biomedicine and Bioinformatics 5th Course - Complex Systems BioMedicine: Molecules, Signals, Networks, Diseases* Catania (Italy) **2009**.

9. J. J Díaz Mochón ORAL COMMUNICATION. High-Throughput Platforms for Transfection Assays. *Eurocombi4*, Firenze (Italy) **2009**.

10. J. J Díaz Mochón, N. Svensen, D. Pouchain, S. Planonh, M. Bradley. POSTER. Nucleic acid encoded peptide libraries. Application to proteases, kinases and cell-based assays. *LabAutomation 2009*, Palm Spring (USA) **2009**.

LANGUAGES:

Spanish: mother tongue

English: fluent, written, read and spoken

Italian: fluent, written, read and spoken

B3 IMPLEMENTATION

B3.1 Quality of host organisation, including adequacy of infrastructures/facilities

The candidate will develop his research and training in the University of Granada. The host institution is fully equipped with all the cutting-edge technology, offering the appropriate equipment, facilities and opportunities to the candidate to carry out top-class research. This organisation counts with history of more than 500 years, and is one of the main Universities in Spain. Today it is internationally recognised at every level: educational, research, cultural and environmental commitment. The University counts with a large number of students PhD students (2200) and undergraduate students (60000). The University of Granada counts with several formation centers committed with knowledge transfer and training in complementary skills (such as Formación Continua, Centro de Lenguas Modernas, Aula Permanente, Centro Mediterráneo y Cursos de Verano). The University of Granada (Spain) maintains analytical service laboratories through which investigators have access among others to a high resolution NMR facility (one 600 MHz, one 500 MHz, one 400 MHz, and one 300 MHz Varian instruments), a MS facility (Bruker MALDITOF, VG Platform II Fisons Instruments spectrometer, and a VG AutoSpec Q high resolution Fisons Instruments mass spectrometer, with capability to perform LC-MS, ESIMS, CI-MS, and EI-MS), and elemental analysis facility. An X-ray diffraction service is also available for the analysis of crystalline samples. The Department of Medicinal and Organic Chemistry at the University of Granada has all equipment needed for carry out the synthetic part of this project. In particular it has a microwave, several fumehood, HPLC and have access to a LC/MS(ESI, APCI, TOF) instrument that has been recently set up on the same building by the University. A new Health-Sciences specialized Campus is under construction, which includes research buildings such as the Biomedical Research Center. This Health Science Technological Park (PTS) is a resource for the creation, implementation and expansion of institutes and companies, which converts knowledge into economic and social development, especially in the Pharmaceutical, Health Sciences, Healthcare and Food industries, making it the first park specialized in healthcare in Spain and one of very few throughout the world. The park has the facilities necessary to offer all the benefits of a modern technological space and aims to integrate the fields of Health Care, Business Development, Research and Teaching with a particular focus on Health and Biomedicine. The PTS is a technological space which helps strengthen the relationships between research and financial sectors, so the university community can make the results of its research as useful and profitable as possible. The combined scientific and technological resources available in the PTS, integrated in Campus of International Excellence BioTic (CEI-BioTic) create an excellent environment for the creation of synergies and offer an ideal location for companies in the biotechnology and health sectors. Located in this PTS is Genyo, a new Pfizer-University of Granada-Junta de Andalucía Centre for Genomics and Oncological Research (GENYO) which has been designed as a space of excellence for research on the genetic basis of diseases, including cancer, as well as on the influence of genetic inheritance in the body's response to certain drugs. Also there it is located Fundación Medina, a non-profit public-private partnership between Merck Sharp and Dohme de España S.A.-Junta de Andalucía, and the University of Granada, which under the Directorship of Dr. Olga Genilloud is focused on the discovery of new compounds and innovative therapies and biotechnologies for unmet medical needs. Within Genyo center, the candidate will be working in collaboration with Dr. Jose Luis Garcia Perez's Research Group, a world-class researcher specialized in reprogramming (see letter of intent of collaboration). In summary, the University of Granada provides a fantastic and world class research environment for my research programme.

Host expertise in training experienced researchers: The European Charter for Researchers has defined standards for early stage researchers with employers ensuring that „all researchers at any stage of their career, regardless of their contractual situation, are given the opportunity for professional development and for improving their employability through access to measures for the continuing development of skills“. The training programme at the University of Granada will be based upon four key elements: the research project (with training on new aspects of research, instrumentation), theoretical training, practical training and generic skills training. The University will provide a range of training and tutoring opportunities in many areas. This includes courses on publication, exploitation, research ethics, research presentations, health and safety. Fellows would be strongly encouraged to participate in areas such as: Spreading Awareness and Knowledge of the Proposed Work *via* the active outreach programmes in Granada that have been internationally recognized; Management of Intellectual Property Rights allowing the effective exploitation of the EU's research base to broaden their generic skill set. The University of Granada has a long-standing

reputation in teaching and training and it has also a demonstrated ability in Marie Curie actions and thus is well set to offer a high-quality training package.

In order to ensure the prospects for professional integration of the candidate, the host institution commits to provide the candidate with the training needed to develop his career as a professional researcher.

Resources will be available to:

- i) Undertake all the required training activities: Candidate's training needs will be identified and specialized courses and workshops will be offered to him to develop his scientific, complementary and organisational skills (i.e.: Specific sessions on European and Spanish legislation, regulation and ethical rules)
- ii) To speed up his research and scientific consolidation allowing focusing not only on bench-work, but most importantly to develop an independent and critical thinking, prepare grant applications and write up the manuscripts for publication
- iii) To attend and/or present his results in National, European and International Meetings. For 2013, participation in four meetings have been foreseen

B3.2 Feasibility and credibility of the project, including work plan

Although being ambitious, the research project is organised in such a way that provides varying degrees of challenge. The more demanding components of the project will, in turn, produce results that have a high impact in the field of study therefore justifying the level of risk associated with them. As such, the research goals outlined are highly feasible and credible. The main concepts are built on prior knowledge and excellence, with this project bringing many things together in a unique and innovative manner. The research plan has an iterative stepwise approach that allows research to progress and be modified and adjusted as it grows. RMS-M has extensive experience on development of microspheres and the applicant can be considered a world-leader in the field of screening platform development in particular in the area of nucleic acid being the inventor of a unique technology. The access to biological samples to exemplify the platforms presented here is guaranteed by the agreed collaboration with Dr. Concha Lopez who is Chief of the Pathology Services and Director of Biobank of cells, tissues and tumours of the Hospital Virgen de las Nieves de Granada. Therefore, if any suggested approach should fail, there are several other routes and avenues to be considered and explored. The alternatives would be getting samples from fully accredited and certified commercial suppliers of human tissues for biomedical research such as American company Origene Technologies Inc., and Europeans Asterand plc. and Cambridge Biosciences Ltd. The project has incorporated a range of achievable goals and the research has several exploitable publishable endpoints. The preliminary work plan and time lines of the research are presented in table shown below. This proposal will be divided in three main areas: (i) synthetic chemistry (ii) platform development and (iii) biological screening. In terms of management the University of Granada will provide the candidate with a well-equipped laboratory space located in the Department of Medicinal and Organic Chemistry where the phase of the project execution involving chemistry techniques will be developed with free use of the equipment that this Department possesses. The applicant will have access to all Nuclear Magnetic Resonance (NMR), mass spectroscopy (MS), microanalysis and x-ray facilities of the University of Granada for the correct identification and characterisation of the structures (expenses details below). The biological samples used for the validation of the proposed technologies will be obtained from Dr. Concha Biobank. Screening will be carried out in the Biological Analysis Unit of the Centre of Scientific Instrumentation of the University of Granada provided with flow cytometry and scanning and microscopy facilities (expenses details below). The preparation of manuscripts to scientific journals and participation in congresses will be done during the whole funding time. The last four months of the last year will include data compilation and reports elaboration. *Justification of request of a research assistant:* The University of Edinburgh has employed the applicant as a Ramon y Cajal Researcher for 5 years (end of contract: 31/05/2016). This job position contemplates teaching duties including lectures, tutorials, lab supervision, student assessments and administration duties. These duties are essential to stabilise the candidate position within the University of Granada. Consequently, the possibility to have a research assistant to support in the lab in a daily basis the candidate research will be of huge help to carry on this research project in an efficient manner. Also, as the field of chemical biology is relatively new in Spain and I believe that area can be strength by bringing high-qualified researches to expand the know-how which will be created during the project. This researcher will also help the candidate to grow his scientific team in Spain by fostering new PhD students who are foreseen to join Diaz Mochon Group.

Diaz Mochon CIG - Timeline											
Milestones	Years										
	Semesters					Months					
	1	3	5	7	9	12	13	15	17	19	21
PNA probes for miRNA detection	1	3	5	7	9	12	13	15	17	19	21
Design of probes against miRNA-21 and 10 (report)											
Synthesis of PNA probes for miRNA detection (for beads and microarray platform) report and materials											
Novel tagged aldehyde modified nucleobases	1	3	5	7	9	12	13	15	17	19	21
Synthesis of four "same-color" fluorescently labelled aldehyde nucleobases (report & materials)											
Synthesis of four color fluorescently labelled aldehyde nucleobases (report & materials)											
Validation of reagents prepared above (GO/NO GO)	1	3	5	7	9	12	13	15	17	19	21
Use of synthetic RNA and MALDI-ToF (report)											
Development of bead based approach* (GO/NO GO)	1	3	5	7	9	12	13	15	17	19	21
Coupling of PNA probes onto microspheres (report & materials)											
Validation by flow cytometry using synthetic RNA (report)											
Development of microarray based approach* (GO/NO GO)	1	3	5	7	9	12	13	15	17	19	21
Coupling of PNA probes onto microarrays (report & materials)											
Validation by fluorescence scanning using synthetic RNA (report)											
Validation of platforms using miRNA coming from biological sources**	1	3	5	7	9	12	13	15	17	19	21
Protocol for miRNA extraction and purification (report)											
Protocol for miRNA profiling using bead-based approach (report)											
Protocol for miRNA profiling using microarray based approach (report)											
Patenting and publication of the works	1	3	5	7	9	12	13	15	17	19	21
Final report											
Annotations											
* One of the two platform might outperformed the other one and at that point a decision will be taken in order to proceed with both or just with one platform											
** In case of not having low enough limit of detection a contingency plan will be applied as described in the text below											

GO/NO GO Point

B3.3 Management: Practical arrangements for the implementation and management of the research project

The applicant will be responsible for the supervision of the project in this application. The chemical part of the project will be developed in the Department of Medicinal Chemistry and Organic Chemistry of the University of Granada. The laboratory located in this department has the necessary infrastructure for the day-to-day running of a synthetic work, several fume cupboards and separated offices with fully network on-line facilities. All technological needs are covered here and also bibliographic consulting and access to online databases such as Beilstein Commander 2000 programs and SciFinder Scholar licence. Necessary reagents in each case are available or easily obtainable. Products from solution synthesis will be purified using suitable techniques (column chromatography with low, high or medium pressure, recrystallization, etc). Characterization of compounds will be carried out by suitable techniques according to the characteristics of each compound. NMR and MS facilities will be used routinely for the characterization of the intermediate and final products obtained during the development of this project. Elemental analysis and X-ray diffraction will help to the determination of the purity and the structure of the final products. LC-MS will be used for the monitoring of reactions progress, characterization of products, and quantification of active products in biological systems (a LC-MS, single quad, Agilent, is available within the Group). Additionally this lab is equipped with all instrument needed to carry out solid phase chemistry.

REQUESTED BUDGET (in euros)	Cost per year	Total cost (4 years)	EC contribution
Contribution to the salary of the applicant	47000	188000	-
Other salary costs (research assistant, part time: 20 hours/week)	12276	49104	49104
TRAVEL COSTS (including registration to conferences)	500	1500	1500
CONSUMABLE COSTS			
Chemicals (resins, chemicals, solvents)	4500	18000	18000
Assays: Cell culture media, culture flasks, microplates, micropipettes plus flow cytometry and microarray related consumables	4224	16896	16896
OVERHEADS	4700	18800	10000
PATENT/PUBLICATION COSTS	1000	4000	4000
TOTAL AMOUNT	25000	296800	100000

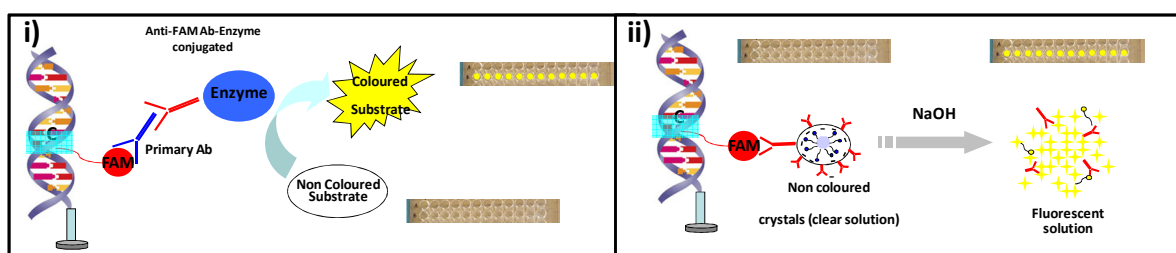
Biological evaluation of the obtained platforms will be done both in the Centre of Scientific Instrumentation at the University of Granada that is equipped with the required techniques for biological evaluation and analysis such as flow cytometry and fluorescent-based equipment, Dr. Concha's Pathology Lab and facilities found at Genyo and Fundacion Medina which are participated by the University of Granada. The international network of the candidate, mainly Prof. Bradley, will also provide required backing if needed.:

- a) Biweekly oral project assessment and meetings with RMS-M and ACL team.
- b) Monthly written reports
- c) Frequent meetings and communication with collaborators.
- d) Quarterly research presentations to the RMSM team and annual presentations to the School of Pharmacy

The University of Granada's Research Transfer Office (OTRI) will administer the intellectual property generated in the course of the work in line with EU recommendations.. Financial management will be carried out by the University of Granada through its Finance Office. Additionally the Department of Medicinal and Organic Chemistry has a full-time Financial Administrator (Mr. Antonio Ortega Arrebola), who will ensure and manage

all payments and track the progress and financial stability of projects in a daily basis.

Contingency plan: The preparation of the synthetic toolkit needed for the project can be considered a very low risk as previously similar molecules have been prepared. Integration of the reagents within the different platform (beads and microarrays) can also be considered low risk due to my previous experience. In this area I rely on the experience of RMS-M for consulting me on the bead based approach. miRNA extraction and handling is not a practical issue due to the large number of protocol which are available. However, the main risk of the project will be the lack of sensitivity of the platform to do direct detection of miRNA both on beads and arrays. In order to address this I have prepared a contingency plan which encompasses two different strategies to amplify the signal by using anti-fluorescein antibodies linked to either (i) enzymes or (ii) FDA crystals (see Figure above). This will increase the sensibility while keeping the high specificity of the technology.



B4 IMPACT

B4.1 Contribution to research excellence by attracting and retaining first class researchers

The integration of the candidate, Dr. Juan J Diaz Mochon, to the EU scientific community will be a great advance to the development of novel platform for biological applications focused on clinical diagnosis. The proposal outlined here show just a fraction of the potential of the new technologies that the candidate can develop within the EU, being able to interconnect areas such as nanoparticles and nucleic acid detection as well as microarrays. The benefits of the successful development of these technologies could ultimately translate our understanding of miRNA biology in cancer into strategies for the control of cancer allowing the EU to gain a major foothold in this emerging area which is undoubtedly now being dominated by the USA. Dr. Diaz Mochon has been a pioneer and emerging leader in the development of technologies to study biological systems while been at the University of Edinburgh, being inventor of an unique technology which allow the identification of nucleic acid without using PCR. Dr. Díaz Mochon has also transferred himself this invention from the bench to a business entity by creating a spin-off company (DestiNA Genomics Ltd.). It is worth mentioning that some of the technologies designed by the candidate are currently under clinical program in collaboration with the National Health System (NHS) of United Kingdom. He has also very developed a novel strategy for the identification of cell-penetrating peptides that allow homing to any cell type or tissue of interest, such as a targeted organ or a tumor. The skills of the candidate related to combinatorial synthesis and of libraries of peptides have been vital for the success of this project. The implementation of high throughput screening approaches such as microarrays technology for clinic diagnosis and treatment of human diseases is an important field of study, and the candidate is in an excellent position to become a team leader in the field, as he has all the required skills to participate and leader its study. Dr. Diaz-Mochon is an excellent team leader as he has quite experience in the training and daily working in several types of personal, from technicians to PhD students and postdoctoral researcher. He is an excellent team player, and has collaborated and collaborates with several laboratories in many countries in the world. He has contributed to the scientific community with excellent studies published in prestigious journals like *Angew. Chem. Int. Ed.*, *Analytical Methods*, *Biomaterials* and *Advanced Materials*. The present application has a tremendous potential to develop lasting co-operations with scientist located in EU and outside the EU. His expertise, skills and potential ensures that Dr. Diaz-Mochon integration to the EU scientific community would be key to increase EU knowledge in the area of molecular diagnosis and prognosis to develop novel tools to fight many severe diseases that nowadays affect to the European citizens due to lack of early diagnosis.

B4.2 Potential and quality of the researcher's long term professional integration in Europe

The University of Granada, in line with the Spanish Ministry of Economy 2008-2011, try to strengthen the incorporation and reintegration of excellence scientists in order to increase its critical mass. Thanks to this effort Dr Diaz Mochon could get offered a five year contract as Ramon y Cajal Researcher which will go from June 2011 to May 2016. During this time, to facilitate the stabilization of the researcher, the candidate will be able to apply for an upgrade that would result in a permanent contract. This lasting engagement will provide the candidate with long-term job stability and will boost the candidate to work towards his independence as researcher and develop new research lines within the host institution, and in doing so contribute to enhance EU scientific excellence. All of these successive positions involve the development of research and teaching and from all of them the applicant can run a research group and be an independent researcher. This way, with his current position the applicant has started an academic and scientific career that will be linked to the Academic environment at the University of Granada. The applicant is willing to become an independent researcher integrated in the scientific community in this University. Thus the consecution of this Marie Curie Career Integration Grant will be of great help as it will allow him to take the first step in this integration process. During the five year-contract, the candidate will be subject to a tracking evaluation process and depending on the scientific results, plus some teaching duties that the candidate will have to perform, obtained during this phase the host institution will commit to create a permanent job for the researcher

B4.3 Potential of transferring knowledge to the host organisation

The candidate integration in the Department of Medicinal and Organic Chemistry of the University of Granada, Spain, will mean the transfer of knowledge of novel chemical-based platforms. This has been agreed by DestiNA Genomics which currently hold the exclusive license of the technology invented by the

candidate (see letter in the Annex). This will make sure that foreground IP based on this previous IP will have operating capacity and will not be blocked. Due to the novel application that the technology proposed here offer it is likely new IP will be created which will remain within the host, keeping the option to be shared with the relevant associated collaborators for instance DestiNA Genomics the University of Edinburgh and the Hospital Virgen de las Nieves in Granada. Within the Univ. of Granada the candidate will be based in the Department of Medicinal and Organic Chemistry which has focused its research in the development of new therapeutic strategies based on the rational design of new selective agents. In recent years this department has established an international reputation in the areas of anticancer therapies and treatment of neurodegenerative diseases. This team of scientists is mostly formed by experts in the field of Medicinal Chemistry, with a focus in the rational design and synthesis of drugs. However development of new biotechnologies capable of doing molecular diagnosis is very important for current drug development programs, which need biomarkers associated to their drugs for tailor different drugs to different genetic profiles of the patient populations. The integration of the candidate in this department will bring very valuable tools to allow more efficient drug designed by the several research groups which conform this department. Moreover the transfer knowledge of this technology from chemistry to biomedical department can be a cornerstone model for the host institution. Making this interaction fluid is key in Life Sciences. It is well established that the potential of the technology proposed here will be key for cancer diagnosis and prognosis which will enhance current cancer treatments.

In the light of the previous results already obtained in which this research project is based, the feasibility of this proposal is high and the outcome very promising, as it will allow provide to clinical researchers new tools for the study miRNA in a way not seen before. In particular the new platforms will shed light in the imminent field of personalised medicine for cancer patients.

B4.4 Capacity to develop lasting co-operation and collaborations with other countries

The candidate has always been engaged in academic collaborations with world-class researchers. Firstly as postdoctoral fellow in the Bradley's research group he has helped Prof. Bradley to disseminate their research and technologies through different areas of research. In July 2008 the candidate was awarded a highly prestigious Fellowship by the Scottish Enterprise (United Kingdom) (£285.000) for a project entitled "Chemical based DNA sequencing and SNP analysis ". From this time onwards he has been developing and planning his own research programme within the School of Chemistry at the University of Edinburgh. This fellowship allowed him to establish many collaborations with a multitude of high-profile researchers within the University of Edinburgh such as Dr. Dhaliwal and Prof Haslett from the Centre for Inflammation Research in the College of Medicine (clinicians in lung inflammation) , Dr. Dave Hay & Prof. John Iredale (MRC Centre for Regenerative Medicine, Edinburgh), Dr. Alison Mackinnon (Centre for Inflammation Research, Edinburgh), Prof. David Porteous (Cancer Research UK Institute, Edinburgh) and Dr. Brickman from the Institute for Stem Cell Research (ISCR). Across Europe collaborations re molecular diagnosis include: Prof. Tim Elliot (CRUK, Southampton), Prof. Salim Kakhoo (Imperial College, London), Prof. Purello from the Department of BioMedicine at the University of Catania (Italy), Prof. Paul J. Coffey (UCM Utrecht, Netherlands) , Dr. Joop Van Helvoort (Center for Translational Molecular Medicine (CTMM) – Netherlands). All these collaborations will keep active if this proposal is granted, as these collaborations are based in the need of development novel chemical strategies to solve the daily problems of these research groups of be able to detect efficiently biomolecules directly related to pathological or biological process. Thus, the establishment of the candidate as an independent researcher team leader will allow the establishment of long-lasting cooperation between the candidate and University of Edinburgh as well as other international Research Centres.

The University of Granada and the University of Edinburgh (UK), candidate's institution, have been collaborating since 2001. It is envisaged this collaboration will continue in the forthcoming years, hastening research projects on molecular diagnosis and high throughput synthesis and screening. In this sense some activities to be carried out within the current project will be developed in collaboration with Prof. Mark Bradey's group (School of Chemistry, University of Edinburgh, United Kingdom).

B4.5 Plans for dissemination and exploitation of results

The research program will be efficient and effectively managed and research progress will be appropriately disseminated within the team and in the scientific community: it is the intention to contribute in high impact factor journals of Medicine, Chemistry, Biological Chemistry and Medicine such as Nature Biotechnology (29.495), ACS Chemical Biology (5.108), J. Biol. Chem. (impact factor 7.368), Angew. Chem. Int Ed. (8.547), J. Med. Chem. (4.134), J. Am. Chem. Soc. (6.025), Chem. Commun. (5.1), Science Translational Medicine (3.5) and Org. Lett. (3.367).

Previous to its publication the potential of the results to be patented will be studied by the office OTRI of the University of Granada.

It is planned as well the annual participation in international congresses with strict evaluation such as the organized annually by prestigious organizations such as the American Chemical Society (ACS) and the Royal Society of Chemistry (RSC).

Dissemination of the new technology and their applications to the general public will also be performed using both mass and social media channels. Dr. Diaz-Mochon has experience in taking part in these PR activities.

This section should include a list of planned dissemination activities, such as publications, conferences, workshops, and websites.

B4.6 Impact of the proposed outreach activities

In order to promote communication between the scientific community and the general public and increase awareness of science, various outreach activities will be carried out. There are various organisations and schemes in Spain that seek to inform society of important and topical issues, such as research with stem cells. There are also opportunities for people to have a more direct involvement in the policy-making process through various public consultation exercises. The University of Granada has a strong commitment to engaging the public in research. In particular The School of Pharmacy website, that provides generic details of current research activities, will be regularly updated to provide details of significant findings generated during the time frame of this fellowship. The University of Granada is also a key partner of the Campus of International Excellence BioTic which mission is to create a strategic partnership composed of public and private players that bring a high level of shared involvement and a sense of shared identity around a common trajectory and model based on a set of differentiating strengths. The candidate will use this environment to create a link between academic and private biotech sectors.

Additionally we will develop our lab website to better describe our work to the lay audience. When appropriate, we will seek interactions with specialist and lay press, as well as opportunities to speak to the public. The candidate has acquired experience in the area of engagement with the public at the University of Edinburgh under the supervision of Elizabeth Stevenson, Outreach/Public Engagement Officer of the School of Chemistry at the University of Edinburgh. The applicant will be able to transfer these skills to the host institution. As a representative example he coordinate a summer school to undergraduates schools to talk with students about Chemistry and attend Open Day at the School to receive students to visit the School and talk to them about research and bring them to the lab and engage with them. Additionally, due to the huge impact of the research that the candidate has developed about "Enzyme-free nucleic acid detection" (published in *Angew. Chemie*), the applicant was engaged with the general public *via* press, radio and national TV, featuring in many of the UK media such as BBC, The Times and The Scotsman due to the potential applications that his technology can offer to the society in terms of facilitating a better treatment. She has been able to explain in lay words how this technology can help to do a more targeted treatment of severe diseases such as cancer.

In order to promote communication between the scientific community and the general public and increase awareness of science, various outreach activities should be outlined in this section. For the planned outreach activities, their *expected impact* should be explained in the proposal. For examples, see box on outreach activities below.

Outreach Activities within Marie Curie Projects

Outreach Activities are dissemination initiatives directed at the general public. The primary goal is to create awareness of the importance of research to society and to raise awareness of Marie Curie Actions. Each applicant is invited to submit an Outreach Activities Plan as part of his/her proposal. The type of outreach activities is freely chosen by the applicant and could range from press articles to exposing students from primary and secondary schools or universities to science, research and innovation in order to develop their motivation to embrace research careers.

Outreach activities and their impact are taken into account during the evaluation of proposals in the light of the principles of the 'European Charter for Researchers' and 'Code of Conduct for the Recruitment of Researchers'. The relevant principle in the Charter is: "**Public engagement** - Researchers should ensure that their research activities are made known to society at large in such a way that they can be understood by non-specialists, thereby improving the public's understanding of science. Direct engagement with the public will help researchers to better understand public interest in priorities for science and technology and also the public's concerns."

Possible outreach activities:

- **Marie Curie Ambassador:** Marie Curie fellows visit schools, universities, community organisations, etc. and promote their research field; Marie Curie fellows - "Ambassadors" - assist teachers in preparing and delivering teaching materials.
- **Workshop Day:** A Marie Curie project runs a workshop/activity day in areas related to the raising of scientific awareness, for school/university students.
- **Summer-School Week:** Students spend one week in a summer school where they receive a first hand experience from the Marie Curie fellows about their current research activities or wider scientific issues; the Marie Curie fellows prepare specific activities, lectures and experiments.
- **Marie Curie Project Open Day:** Students and the general public visit the research institutions or labs and receive a first hand experience or lectures.
- **Public talks, TV-Talks, podcasts and articles in Newspapers:** Marie Curie fellows give a public talk/TV interview or write an article in the local newspaper about the results of the project and how these results could be relevant to the general public.
- **e-Newsletters:** Marie Curie fellows develop a web-based document to be released on internet to the attention of the public at large (e.g. Wikipedia).
- **Multimedia releases:** Marie Curie fellows make video-clips to be released on internet, in spaces open to the public at large.

B5 ETHICAL ISSUES

Ethics is central to scientific integrity, honesty and clarity of science. It is considered essential by the REA and the European Commission in the research activities that it funds or carries out itself. This means that in any proposal submitted to the 7th Framework programme, ethics issues must be identified and addressed. Proposals that pose ethics concerns will be flagged. If some aspects are incomplete, clarification may be sought, but this will cause delays in the application process.

Considering **ethics issues** from the concept stage of a proposal enhances the quality of research. Applicants should take time to consider the benefit/burden balance of the research activities; consider the impact of the research, not only in terms of scientific advancement, but also in terms of

Research on Human Embryo/ Foetus			YES	Page
*	Does the proposed research involve human Embryos?		NO	
*	Does the proposed research involve human Foetal Tissues/ Cells?		NO	
*	Does the proposed research involve human Embryonic Stem Cells		NO	

	(hESCs)?		
*	Does the proposed research on human Embryonic Stem Cells involve cells in culture?	NO	
*	Does the proposed research on Human Embryonic Stem Cells involve the derivation of cells from Embryos?	NO	
	I CONFIRM THAT NONE OF THE ABOVE ISSUES APPLY TO MY PROPOSAL	YES	

Research on Humans		YES	Page
*	Does the proposed research involve children?	NO	
*	Does the proposed research involve patients?	NO	
*	Does the proposed research involve persons not able to give consent?	NO	
*	Does the proposed research involve adult healthy volunteers?	NO	
	Does the proposed research involve Human genetic material?	NO	
	Does the proposed research involve Human biological samples?	YES	6*
	Does the proposed research involve Human data collection?	NO	
	I CONFIRM THAT NONE OF THE ABOVE ISSUES APPLY TO MY PROPOSAL	YES	

Privacy		YES	Page
	Does the proposed research involve processing of genetic information or personal data (e.g. health, sexual lifestyle, ethnicity, political opinion, religious or philosophical conviction)?	NO	
	Does the proposed research involve tracking the location or observation of people?	NO	
	I CONFIRM THAT NONE OF THE ABOVE ISSUES APPLY TO MY PROPOSAL	NO	

Research on Animals		YES	Page
	Does the proposed research involve research on animals?	YES	9**
	Are those animals transgenic small laboratory animals?	NO	
	Are those animals transgenic farm animals?	NO	
*	Are those animals non-human primates?	NO	
	Are those animals cloned farm animals?	NO	
	I CONFIRM THAT NONE OF THE ABOVE ISSUES APPLY TO MY PROPOSAL	YES	

Research Involving Developing Countries		YES	Page
	Does the proposed research involve the use of local resources (genetic, animal, plant, etc)?	NO	
	Is the proposed research of benefit to local communities (e.g. capacity building, access to healthcare, education, etc)?	NO	
	I CONFIRM THAT NONE OF THE ABOVE ISSUES APPLY TO MY PROPOSAL	NO	

Dual Use		YES	Page
	Research having direct military use	NO	
	Research having the potential for terrorist abuse	NO	

I CONFIRM THAT NONE OF THE ABOVE ISSUES APPLY TO MY PROPOSAL	YES	
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Consistency with part A		YES	NO
I CONFIRM THAT THE INFORMATION GIVEN IN THIS TABLE IS CONSISTENT WITH THE INFORMATION PROVIDED ON ETHICS IN PART A, PAGE A1	YES	YES	n/a

* See comments in Page 6 and Annex 1, 4 and 5

** The project will use biospecimen, blood, from mice to validate the platforms to be developed. These samples will thus be used to analysis their miRNA levels and nothing else. These biospecimens will be purchased by a commercial source such as The Jackson Laboratory which is accredited by the Association for Assessment and Accreditation of Laboratory Animal Care International (AAALAC International) and also have an Animal Welfare Assurance with the Office of Laboratory Animal Welfare (OLAW). Therefore the project itself does not include animal breeding activities.

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ENDPAGE

**PEOPLE
MARIE CURIE ACTIONS**

Marie Curie CareerIntegration Grants

Call: FP7-PEOPLE-2012-CIG

PART B

“CHEMiRNA”

ANNEX:

1. Letter from Dr. Angel Concha Lopez
2. Letter from Dr. Jose Luis Garcia Perez
3. Letter from CEO's DestiNA Genomics. Ltd
4. Collaboration Agreement with Dr. Angel Concha Lopez
5. Registration of the project with the Local Ethic Committee for its Ethical Approval Evaluation